

1 Intersections of curves

1.1 Resultants

Lemma 1.1.1. For R a UFD, and $f, g \in R[x]$ of degree $d \geq 1$ and $e \geq 1$ respectively, f and g have a non-constant common factor iff there exists non-zero polynomials $a, b \in R[x]$ such that:

$$af + bg = 0$$

where $\deg(a) \leq e - 1$ and $\deg(b) \leq d - 1$.

Proof. Suppose f, g have a common factor h of degree $n \geq 1$, then there exists $p, q \in R[x]$ such that $f = ph$ and $g = qh$, so we can have:

$$qf - pg = 0$$

where $\deg(q) = e - n \leq e - 1$ and $\deg(p) = d - n \leq d - 1$.

If, instead, we can find such $a, b \in R[x]$ then be factoring af into irreducible components:

$$af = c \cdot h_1^{\alpha_1} h_2^{\alpha_2} \cdots h_k^{\alpha_k} = -bg$$

if f and g contain none of the same factors, g must divide a , which contradicts the degree requirement. $\square$

This tells us that the polynomials f, g are coprime iff the polynomials:

$$x^{e-1}f, \dots, xf, f, x^{d-1}g, \dots, xg, g$$

are linearly independent. The best test we know for linear independence is the determinant, leading us to the definition of the resultant.

Definition 1.1.2. For R a UFD, and $f, g \in R[x]$ of the form:

$$f(x) = a_d x^d + \cdots + a_1 x + a_0 \quad g(x) = b_e x^e + \cdots + b_1 x + b_0$$

where $a_d, b_e \neq 0$, the **resultant** of f, g is:

$$\mathcal{R}_{fg} := \begin{vmatrix} a_d & a_{d-1} & a_{d-2} & \cdots & a_0 & & & & & \\ & a_d & a_{d-1} & \cdots & a_1 & a_0 & & & & \\ & & a_d & \cdots & a_2 & a_1 & a_0 & & & \\ & & & \ddots & & \ddots & \ddots & \ddots & & \\ & & & & a_d & a_{d-1} & a_{d-2} & \cdots & a_0 & \\ b_e & b_{e-1} & b_{e-2} & \cdots & b_1 & b_0 & & & & \\ & b_e & b_{e-1} & \cdots & b_2 & b_1 & b_0 & & & \\ & & b_e & \cdots & b_3 & b_2 & b_1 & b_0 & & \\ & & & \ddots & & \ddots & \ddots & \ddots & \ddots & \\ & & & & b_e & b_{e-1} & \cdots & b_2 & b_1 & b_0 \end{vmatrix} \begin{matrix} \left. \vphantom{\begin{matrix} a_d & a_{d-1} & a_{d-2} & \cdots & a_0 \\ & a_d & a_{d-1} & \cdots & a_1 & a_0 \\ & & a_d & \cdots & a_2 & a_1 & a_0 \\ & & & \ddots & & \ddots & \ddots & \ddots \end{matrix}} \right\} e \text{ rows} \\ \left. \vphantom{\begin{matrix} a_d & a_{d-1} & a_{d-2} & \cdots & a_0 \\ & a_d & a_{d-1} & \cdots & a_1 & a_0 \\ & & a_d & \cdots & a_2 & a_1 & a_0 \\ & & & \ddots & & \ddots & \ddots & \ddots \end{matrix}} \right\} d \text{ rows} \end{matrix}$$

the determinant of this $(d + e) \times (d + e)$ **Sylvester matrix**.

Corollary 1.1.3. f and g share a non-constant common factor iff $\mathcal{R}_{f,g} = 0$.

Proposition 1.1.4. If $\text{Frac}(R)$ has characteristic zero, then $f \in R[x]$ has a repeated factor iff $\mathcal{R}_{f,f'} = 0$.

Proof. The product rule for differentiation gives $(\Rightarrow)$, if f and f' share a common factor g , then $f = gh$ and $f' = gh' + g'h$ being a multiple of g implies $g \mid h$, thus $g^2 \mid f$. $\square$

If f is the quadratic $ax^2 + bx + c$, then the resultant $\mathcal{R}_{f,f'}$ is $-a(b^2 - 4ac)$, so we recover the normal condition that the discriminant be nonzero.

1.2 Weak Bézout

Throughout this section, we want to consider two projective plane curves:

$$C = \mathbb{V}(F(x, y, z)) \quad C' = \mathbb{V}(G(x, y, z))$$

for homogeneous polynomials $F, G \in k[x, y, z]$, which we view as $R[x]$ over the UFD $R = k[y, z]$, given by:

$$\begin{aligned} F(x, y, z) &= f_0(y, z)x^d + f_1(y, z)x^{d-1} + \cdots + f_d(y, z) \\ G(x, y, z) &= g_0(y, z)x^e + g_1(y, z)x^{e-1} + \cdots + g_e(y, z) \end{aligned} \tag{1}$$

We know these curves won't cover all of $\mathbb{P}^2$, as otherwise for all $c \in k$, the union $C \cup C' = \mathbb{V}(FG)$ would contain the line $\{x - cy\} = 0$, so this must be a factor of the product, which contradicts having finite degree. So our curves avoid some common point; by applying a projective transformation, we may assume that they both miss $[1, 0, 0]$.

Proposition 1.2.1. There exist polynomials $A, B \in k[x, y, z]$ with x -degrees less than e, d respectively such that:

$$\mathcal{R}_{F,G}(y, z) = AF + BG \in k[y, z]$$

Furthermore, if $\mathcal{R}_{F,G}$ is nonzero, it will be a homogeneous polynomial of degree de .

Proof. We will perform a series of column operators on the Sylvester matrix: for each $1 \leq k \leq d + e - 1$, add x^{d+e-k} times the k th column to the last column. This doesn't effect the determinant, and gives the final column the form:

$$(x^{e-1}F, \dots, xF, F, x^{d-1}G, \dots, xG, G)$$

we can now expand the determinant along this column, giving the polynomials A, B ; as every other entry in the matrix is in $k[y, z]$ the degrees of A and B are as desired.

Now to show $\mathcal{R}_{F,G}$ is homogeneous, we can write:

$$\mathcal{R}_{F,G} = \det(-) = \sum_{\sigma \in S^{d+e}} (-1)^\sigma \prod_{i=1}^d f_{\sigma(i)-i} \prod_{j=1}^e g_{\sigma(j+e)-j}$$

each of these is a product of homogeneous polynomials of degree:

$$\begin{aligned} \sum_{i=1}^d \deg(f_{\sigma(i)-i}) + \sum_{j=1}^e \deg(g_{\sigma(j+e)-j}) &= \sum_{i=1}^d \sigma(i) - i + \sum_{j=1}^e \sigma(j+e) - j \\ &= \frac{(d+e)(d+e+1)}{2} - \frac{d(d+1)}{2} - \frac{e(e+1)}{2} = de \end{aligned} \quad \square$$

Proposition 1.2.2. $\mathcal{R}_{F,G}(y, z)$ is zero iff F and G share a non-constant common factor in $k[x, y, z]$.

Proof. We already know $\mathcal{R}_{F,G}$ is zero iff F and G share a non-constant common factor in $k[x, y, z]$ with positive x -degree. If F and G share a non-constant common factor in $k[y, z]$, then this contradicts the assumption that F and G are nonzero on $(1, 0, 0)$, so all non-constant factors have positive x -degree. $\square$

Our assumption that we can apply a projective transformation such that C and C' both miss $[1, 0, 0]$ is the same as F and G being nonzero on $(1, 0, 0)$.

Theorem 1.2.3 (Weak Bézout theorem). The intersection $C \cap C'$ is nonempty, and either $|C \cap C'| \leq de$ or $C \cap C'$ contains a plane curve.

Proof. Consider $\mathcal{R}_{F,G} \in k[y, z]$ which is either zero or homogeneous of degree de . If it is zero, then by the previous proposition our polynomials have a non-constant common factor H , which corresponds to $C \cap C'$ containing the plane curve $\{H = 0\}$.

If $\mathcal{R}_{F,G}$ is not zero, as we are working over an algebraically closed k , we can factor this as

$$\mathcal{R}_{F,G}(y, z) = \prod_{i=1}^{de} (a_i y + b_i z)$$

where at any fixed (y_0, z_0) , $F(x, y_0, z_0)$ and $G(x, y_0, z_0)$ have a common root at some x_0 iff $\mathcal{R}_{F,G}(y_0, z_0) = 0$, which is equivalent to finding some i with $a_i y_0 + b_i z_0 = 0$. So there is an $x_0 \in k$ with $[x_0, y_0, z_0] \in C \cap C'$ iff there is some i with $[y_0, z_0] = [b_i, -a_i]$, thus $C \cap C'$ is nonempty.

Now we just have to bound $|C \cap C'|$, suppose there exists some set $S \subseteq C \cap C'$ of $de + 1$ distinct points. Then, by the same argument to show $C \cup C' \subsetneq \mathbb{P}^2$, the set of projective lines between any pair of points in S can be assumed to miss $[1, 0, 0]$. Now for all $[x, y, z] \in C \cap C'$, we have $(y, z) \neq (0, 0)$ and $a_i y + b_i z = 0$ for some $1 \leq i \leq de$, so for S to exist there must be two distinct $[x, y, z], [x', y', z']z \in S$ corresponding to the same i , but then the unique line through these points is $\{a_i y + b_i z = 0\} \ni [1, 0, 0]$, a contradiction. $\square$

To get the equality of the full Bézout's theorem, we need to count our points geometrically, instead of set-theoretically, this is called an intersection multiplicity. First we see a few applications of what we've already developed.

Proposition 1.2.4. Nonsingular plane curves are irreducible, and irreducible plane curves have at most finitely many singular points.

Proof. Let C be the reducible plane curve given by $P(x, y, z)Q(x, y, z) = 0$, we know P and Q have an intersection, at this point the derivative of PQ is also 0.

Now if C' is the irreducible plane curve $F(x, y, z) = 0$, wlog we may assume $F_x \neq 0$, then as F is irreducible, F and F_x don't share any non-constant common factors. All singular points of C satisfy $F = F_x = 0$, which by Bézout's theorem is a finite set. $\square$

1.3 Conics

A conic is a projective plane curve of degree 2, the classification of conics of $\mathbb{R}$ are the ellipses, parabolas, hyperbolas, and degenerate cases. Over $\mathbb{C}$ they are all equivalent.

Theorem 1.3.1. If C is an irreducible conic over an algebraically closed field of characteristic zero, then C is projectively equivalent to $\{x^2 = yz\}$.

Proof. As all but finitely many points of C are singular, we may take $p \in C$ smooth and $q \in T_p C$ by applying a projective transformation we may assume $p = [1, 0, 0]$ and $q = [0, 1, 0]$ so $T_{[1,0,0]}C = \{z = 0\}$. If C is defined by $F(x, y, z) = ax^2 + bxy + cy^2 + dyz + ez^2 + fzx = 0$, then we know: $[1, 0, 0] \in C \implies a = 0$, $T_{[1,0,0]}C = \{z = 0\} \implies b = 0$, and F irreducible $\implies c, d \neq 0$; which leaves F in the form

$$F(x, y, z) = (\sqrt{c}y)^2 + (dx + ey + fz)z$$

so C is equivalent, by projective transformation, to $\{x^2 - yz\} = 0$. $\square$

Corollary 1.3.2. A conic over an algebraically closed field of characteristic 0 is smooth iff it's irreducible.

Proposition 1.3.3. Every five points in $\mathbb{P}^2$ lie on a conic, if no four of the points are collinear, then this conic is unique.

Proof. If we write the defining equation of the conic explicitly as in the previous proof, then the five points on C determine a system of five linear equations on the 6 coefficients of F , this will admit a nonzero solution.

Suppose five points lie on two different conics, C and C' , then Bézout's theorem says their intersection must contain some irreducible curve C_0 which must be linear for $C \neq C'$. It is an exercise to show that C and C' can now be seen as the union of C_0 with lines $L \neq L'$. If at most three of our points lie on C_0 , then at least two lie on both L and L' , but as the line between two points is unique, this is a contradiction. Thus at least four of the points lie on C_0 , so are collinear. $\square$

1.4 Intersection multiplicities and strong Bézout

Definition 1.4.1. For plane curves C, D and $p \in \mathbb{P}^2$ the **intersection multiplicity** $I_p(C, D)$ of C and D at p satisfies the following:

1. $I_p(C, D) = I_p(D, C)$;
2. if $p \notin C \cap D$, then $I_p(C, D) = 0$;

3. if $C \cap D$ contains the irreducible curve C_0 and $p \in C_0$, then $I_p(C, D) = \infty$;
4. if C and D are distinct lines intersecting at p , then $I_p(C, D) = 1$;
5. if $D = D_1 \cup D_2$ is a union of plane curves, then $I_p(C, D) = I_p(C, D_1) + I_p(C, D_2)$;
6. if C and D are defined by $F(x, y, z) = 0$ and $G(x, y, z) = 0$ respectively, and if D' is given by $FQ + G = 0$ for some homogeneous Q , then $I_p(C, D) = I_p(C, D')$.

To show the existence of an intersection multiplicity, we'll need a few more results about the resultant.

Proposition 1.4.2. For R a UFD, and $f, g \in R[x]$ non-constant polynomials which factor into linear terms:

$$\begin{aligned} f(x) &= a(x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_d) \\ g(x) &= b(x - \mu_1)(x - \mu_2) \cdots (x - \mu_e) \end{aligned}$$

then their resultant is

$$\mathcal{R}_{f,g} = a^e b^d \prod_{\substack{1 \leq i \leq d \\ 1 \leq j \leq e}} (\lambda_i - \mu_j)$$

Proof. We can assume f and g are monic, then each of the coefficients a_k and b_k are homogeneous polynomials in the λ_i and μ_j of degree $n - k$ and $m - k$ respectively. So the ij th entrance of the Sylvester matrix is a degree $n + i - k$ if $1 \leq i \leq m$, or $i - j$ if $m + 1 \leq i \leq n + m$ homogeneous polynomial, and so by the same argument as in Prop 1.2.1, the resultant is a homogeneous degree de polynomial in the $d + e$ variables which vanished precisely when one of the $\lambda_i = \mu_j$, thus is a scalar multiple of our desiderata. If we send all the λ_i to 1 and μ_i to 0, we get a homomorphism from our big polynomial ring to $R[x]$, which sends our leading coefficient to $\mathcal{R}_{(x-1)^d, x^e}$, it is a straightforward computation to show this is 1. $\square$

Corollary 1.4.3. For all $c \in R$, and f as above, $\mathcal{R}_{x-c, f} = f(c)$.

For this case, we don't actually need to assume f splits into linear factors in R , if we do all the work in an algebraic closure of the field of fractions of R .

Corollary 1.4.4. For R a UFD and $f, g, h \in R[x]$ non-constant polynomials, then $\mathcal{R}_{f, gh} = \mathcal{R}_{f, g} \mathcal{R}_{f, h}$.

Proof. We also work in k and algebraic closure of the field of fractions of R . Here we can write

$$\mathcal{R}_{f, gh} = \prod_{i=1}^d (gh)(\lambda_i) = \prod_{i=1}^d g(\lambda_i) \prod_{i=1}^d h(\lambda_i) = \mathcal{R}_{f, g} \mathcal{R}_{f, h} \quad \square$$

Corollary 1.4.5. If, as above, f, g , and $hf + g$ have positive degree and f is monic, then $\mathcal{R}_{f, g} = \mathcal{R}_{f, hf + g}$.

Proof. Same structure, it can be an exercise. $\square$

We're now prepared to prove half of the proposition.

Theorem 1.4.6. As in the definition, there is a unique intersection multiplicity $I_p(C, D)$, which is defined to satisfy 2 and 3, and otherwise is given by the following.

Let $C' \subset C$ and $D' \subset D$ be the unions of components not common to C and D , with a projective transformation applied such that $[1, 0, 0] \notin C', D'$, or any of the lines through pairs of points of $C' \cap D'$. If C' and D' are given by $\{F(x, y, z) = 0\}$ and $\{G(x, y, z) = 0\}$ respectively, and $p = [a, b, c]$, then $I_p(C, D)$ is the multiplicity of $[b, c]$ as a root of $\mathcal{R}_{F, G}(y, z)$.

Proof. The proof of uniqueness is Theorem 3.18 of Kirwan's book: the rough sketch is that once we've shown $I_p(C, D) = k$, we use the six axioms to express $I_p(C, D)$ as a combination of intersection multiplicities strictly less than k and uniqueness follows by induction on k .

Properties 1, 2, 3, 5, 6 are all immediate from the definition of the resultant or the previous propositions, so all that's left is 4, which is just a direct computation. $\square$

Theorem 1.4.7 (Bézout's theorem). For k an algebraically closed field and C, D plane curves of degrees $d, e \geq 1$ respectively with no common components, then

$$\sum_{p \in C \cap D} I_p(C, D) = de$$

Proof. We follow the normalisation procedure in the proof of the weak Bézout's theorem and get a bijection between points $p = [a, b, c] \in C \cap D$ and roots $[b, c]$ of $\mathcal{R}_{F,G}(y, z)$, now the multiplicity of this root is $I_p(C, D)$ and $\mathcal{R}_{F,G}(y, z)$ has exactly de roots, so we are done. $\square$

Points p in the intersection $C \cap D$ with $I_p(C, D) = 1$ are called **transverse intersections**.

Proposition 1.4.8. If k is algebraically closed, and p is a smooth point on the projective plane curves $C = \{F = 0\}$ and $D = \{G = 0\}$, then $I_p(C, D) = 1$ iff $T_p C$ and $T_p D$ are different.

Proof. Wlog assume $C \neq D$, both are irreducible, and $\deg(C) \geq \deg(D) \geq 1$. Now the intersection $C \cap D$ is finite, and we can apply a projective transformation such that $p = [0, 0, 1]$, $T_p D = \{x = 0\}$, and none of C , D , or the lines through pairs of points in $C \cap D$ pass through $[1, 0, 0]$. $I_p(C, D)$ is now the greatest multiplicity y^k which divides $\mathcal{R}_{F,G}(y, z)$ $\square$

1.5 Cubics

Theorem 1.5.1 (Cayley-Bacharach). If C and D are cubics that intersect at 9 distinct points, and C' is any cubic that passes through 8 of these points, then C' belongs to the pencil of cubics

$$\{C_{[a,b]} = \{aF + bG = 0\} \mid [a, b] \in \mathbb{P}^1\}$$

and so also contains the 9th point.

Theorem 1.5.2 (Pascal's theorem). For all sets of six points $\{p_1, p_2, p_3, q_1, q_2, q_3\}$ on a conic C , no three of which are collinear, if L_{ij} is the line through p_i and q_j then for the points $r_i = L_{jk} \cap L_{kj}$, for all $\{i, j, k\} = \{1, 2, 3\}$, are collinear.

Corollary 1.5.3 (Hexagrammum mysticum).

Theorem 1.5.4. If C is a smooth cubic over an algebraically closed field of characteristic zero, the C is projectively equivalent to $y^2z = x(x - z)(x - \lambda z)$ for some $\lambda \neq 0, 1$.

Theorem 1.5.5. A smooth cubic curve over an algebraically closed field of characteristic zero has exactly nine inflection points.

2 Genus